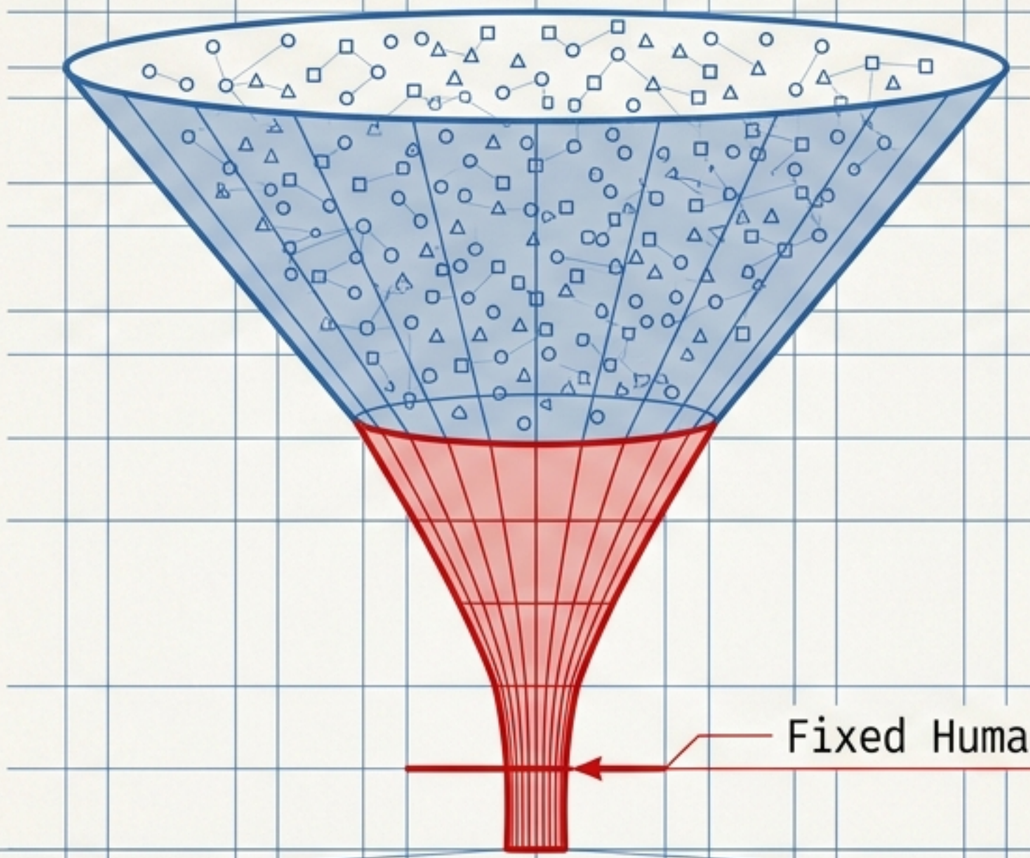
Automating Ideation and the Limits of Formal Logic

A Blueprint for Corpus Recombination, Obligated Composition, and the Hazards of Machine Scale

System Architecture &
Decision Matrix // Version 1.0

Gemini Notebook

The Combinatorics of the Unwritten



$n = 488$ eligible works

$\binom{n}{3} = n(n-1)(n-2)/6$

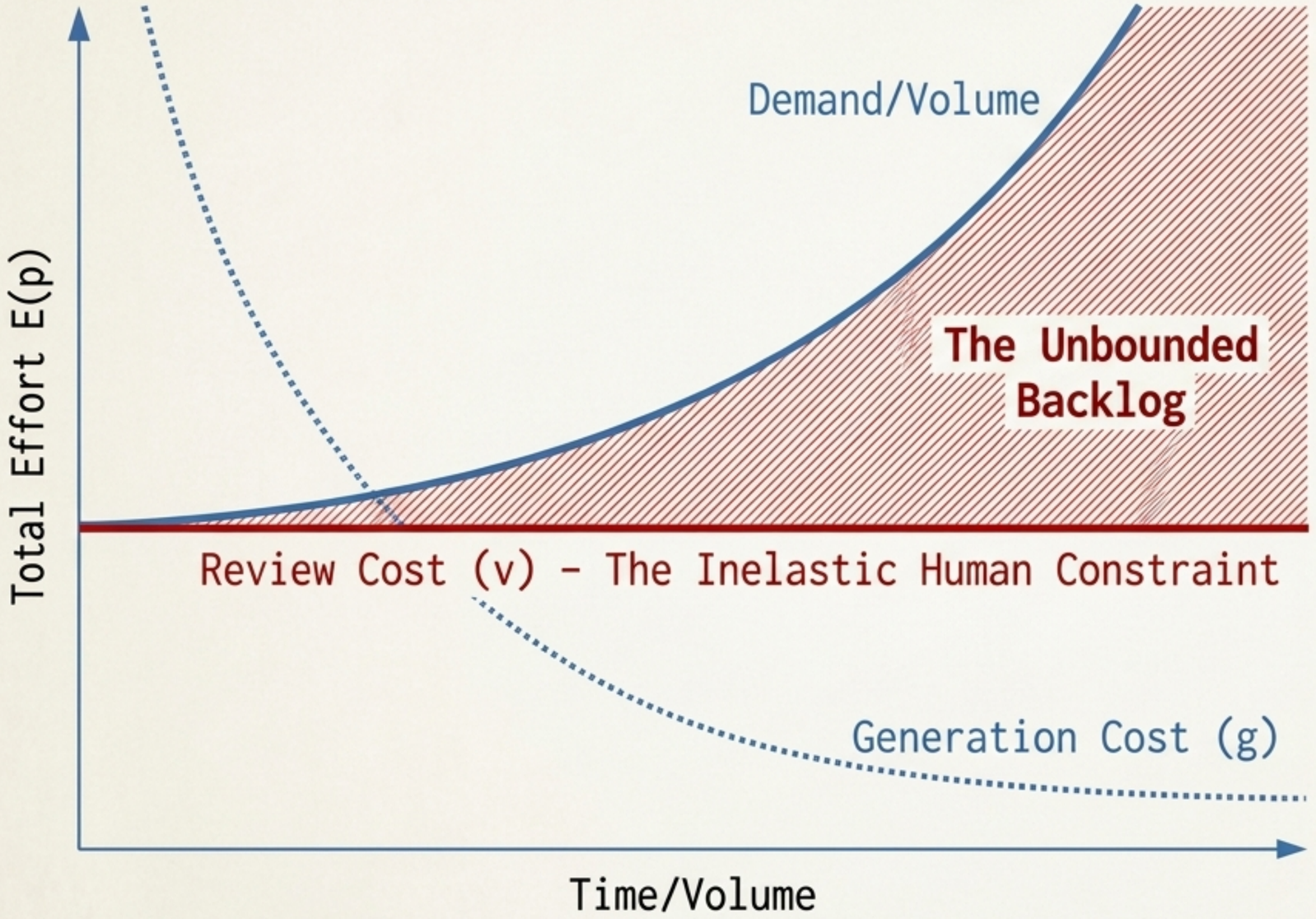
Total: 19,250,136 combinations

Fixed Human Review Capacity (s)

The combinatorial arithmetic proves that the supply of juxtapositions is effectively inexhaustible. The limiting resource is the strict, unyielding work of reading, testing, and writing the argument that survives.

Gemini Notebook

The Jevons Paradox in Automated Ideation



Callout 1:

Price = g + v

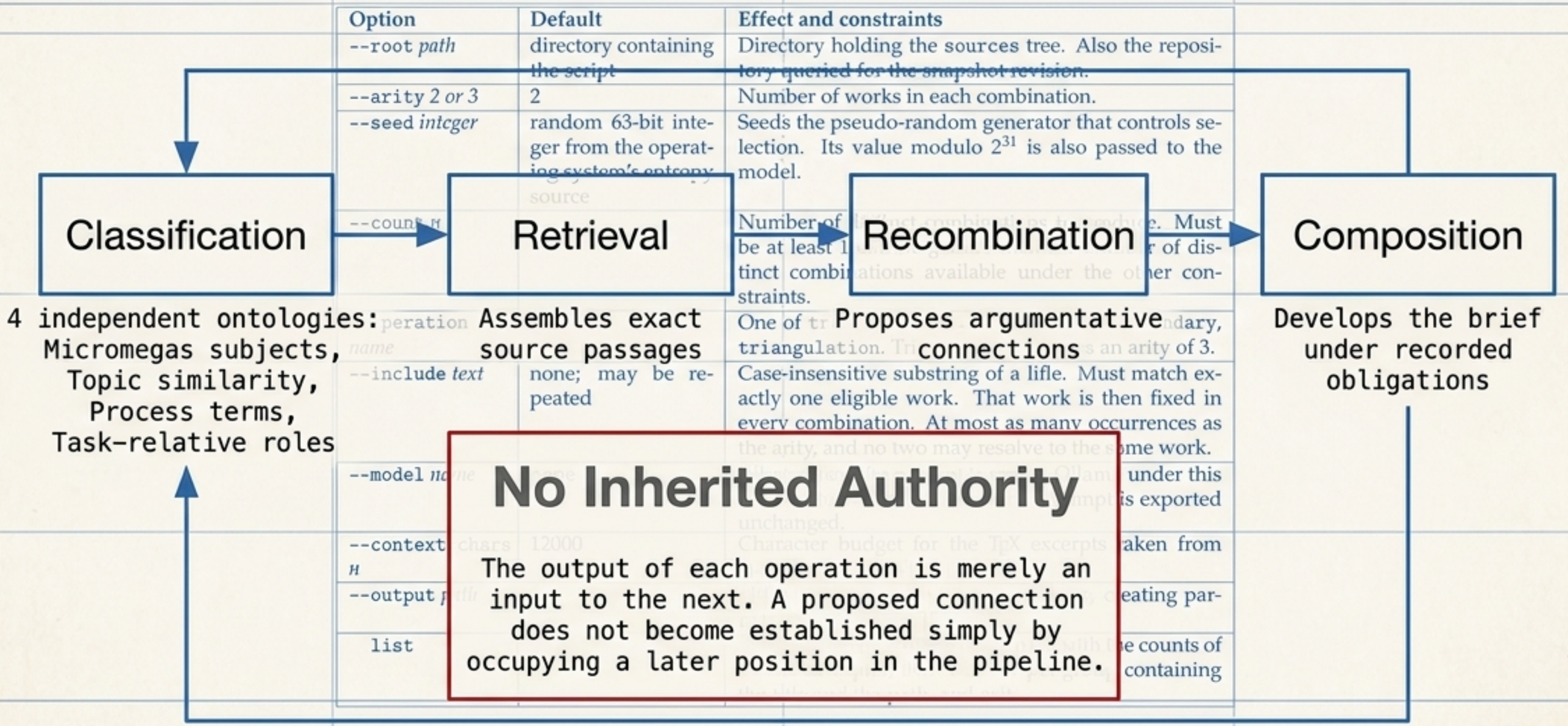
When generation (g) approaches zero, review burden (v) rises by factor κ.

Callout 2:

An author at capacity pushed by any efficiency improvement accumulates unreviewed briefs at a linear, unbounded rate.

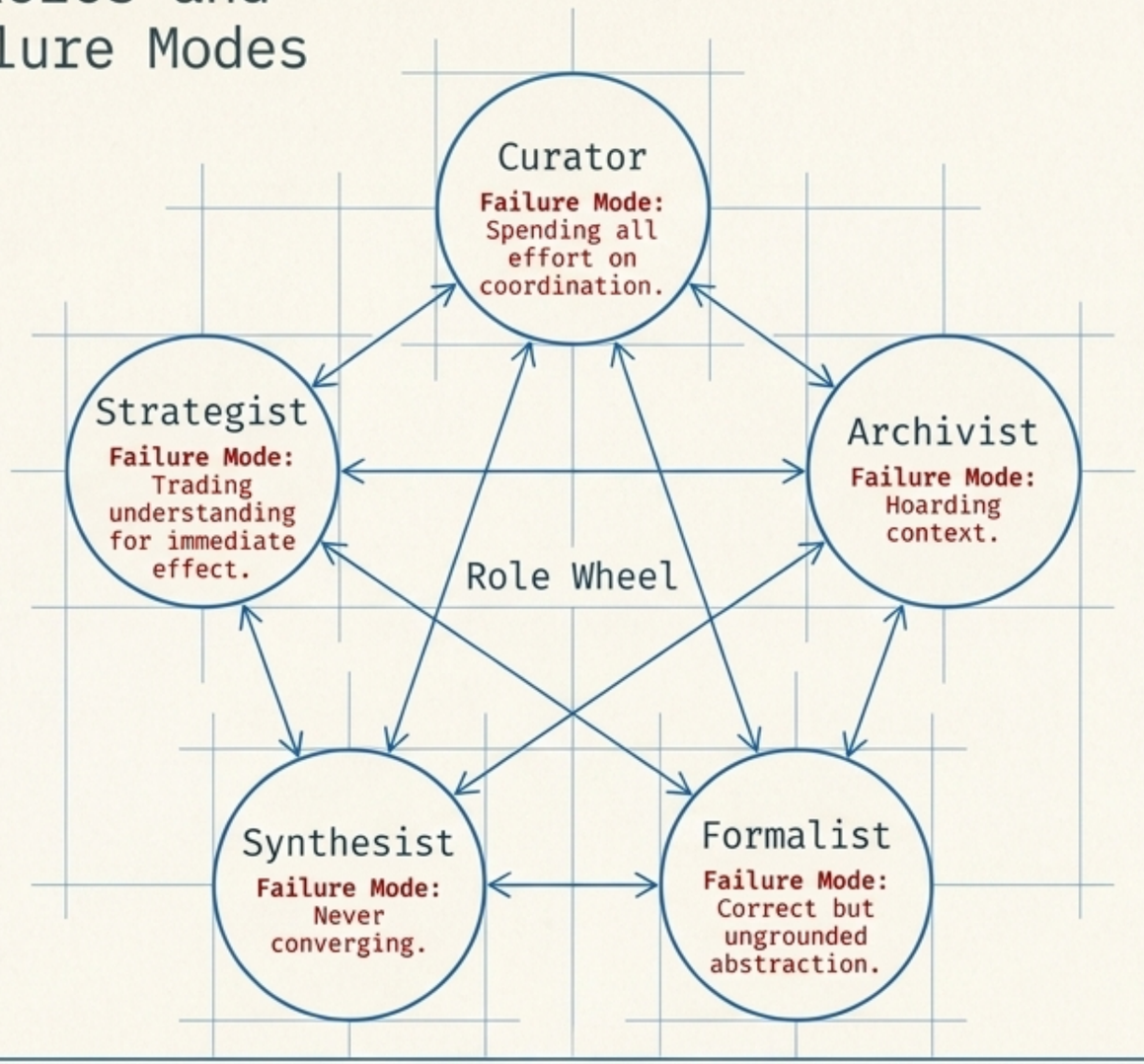
Gemini Notebook

The Yarnwinder Pipeline: Constraining the Corpus



Gemini Notebook

Task-Relative AI Roles and their Expected Failure Modes



The failure modes are the useful part. Roles are queries put to passages; they own nothing and carry no evidential authority.

Gemini Notebook

The Collapse of Metrics Under Optimization

Extremal Gaming	Regressional Gaming	Adversarial Gaming
Maximizing a proxy destroys the target.	Selection on noisy proxies mathematically overstates actual quality.	Self-certification.
The Unigram Shannon Entropy score formula min(100, max(0, 100-10H)). Because it is permutation invariant, minimizing entropy to cheat the score yields a single repeated token. (e.g., A shuffled 13-word string retains 1.925 nats).	If briefs are selected because a heuristic scored them highly, examination will routinely find them less good than the score indicated.	A model that drafts a section and simultaneously proposes that its own obligations are discharged is optimizing its own score via the cheapest route.

Gemini Notebook

Rejecting Scores for the Formal Ledger of Obligations

Blended Numerical Scores

- Requires a model to evaluate text it just wrote.
- Converts failures to parse into default scores.
- Rewards metric gaming and repetition.

The Ledger of Obligations

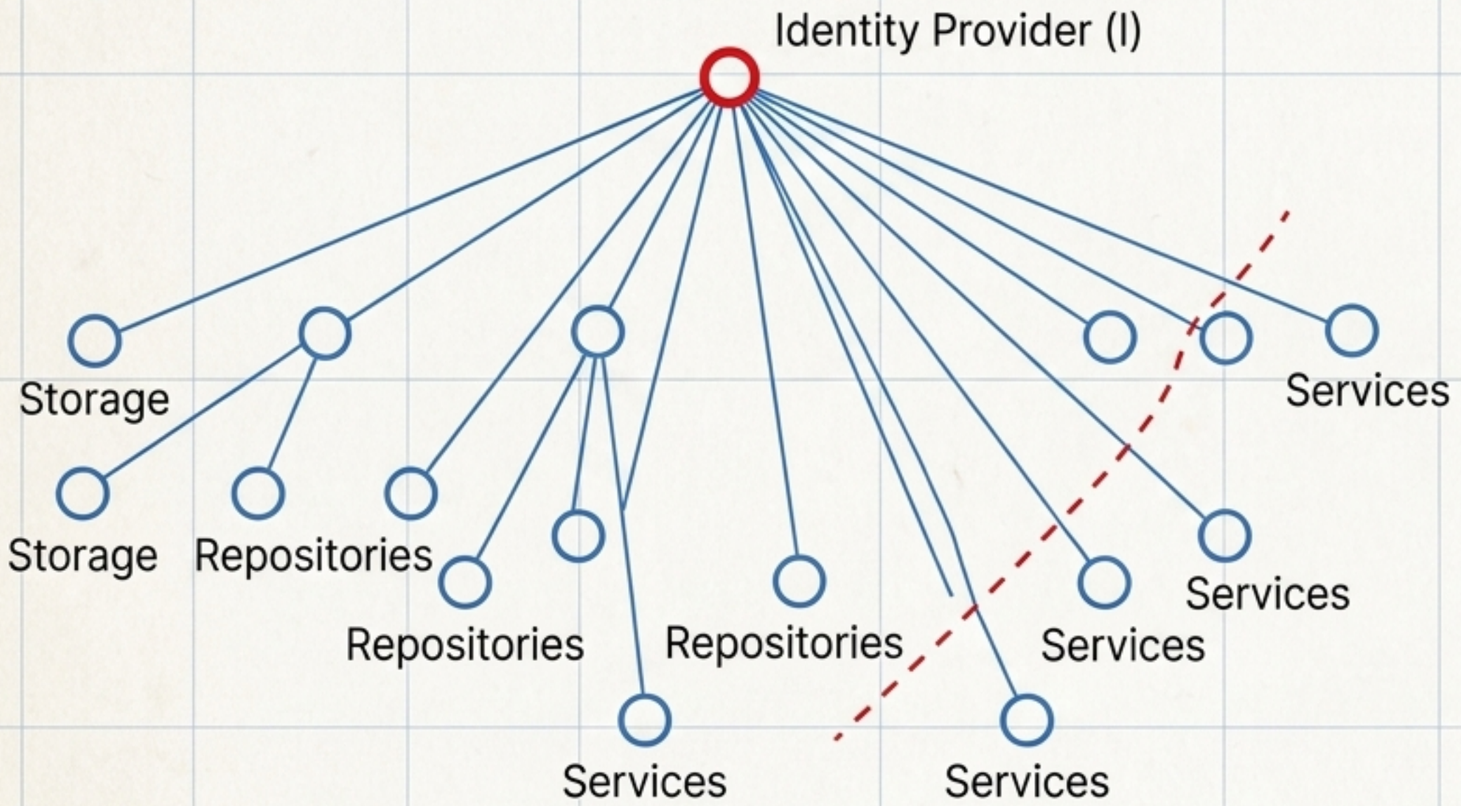
- [ ] 1. Source interpretations are accurate within claimed scope.
- [ ] 2. Connecting premises are explicitly explained, not implied.
- [ ] 3. Material objections receive answers, concessions, or scope restrictions.
- [ ] 4. Open claims remain explicitly listed.

Core Logic Rule: $\Gamma \vdash \text{Confirmed } o$

A model's assertion is derivable for every obligation and costs nothing.
Confirmation is derivable only when a human reader supplies it.

Gemini Notebook

Structural Vulnerability: The Illusion of “Small Footprint”



The Theorem

$$Acc(\{I\}) \cap (\{I\} \cup Dep(I)) = \emptyset$$

Loss of an identity provider propagates to exactly the dependents fixed by the sign-in structure.

The Fallacy

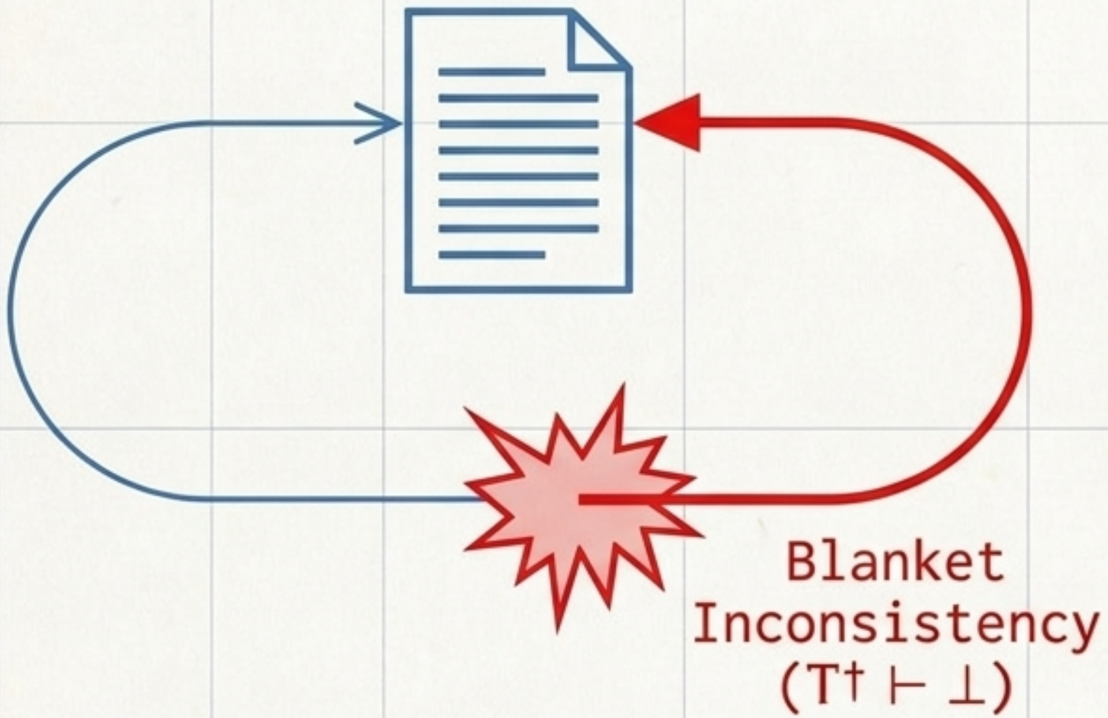
Provider exposure is determined strictly by structural dependency, not by the volume of data held or the revenue derived (the footprint).

The Antidote

Independent sign-in routes and replicas ($M(v) \subseteq R$) structurally remove the dependence, severing the red lines in the diagram.

Gemini Notebook

The Recursion of Self-Endorsement



The Temptation:

Adopting an axiom that states "whatever this development proves is warranted" ($Prov(\ulcorner H \urcorner) \rightarrow H$).

Löb's Theorem:

A theory can only establish its own soundness for sentences it already proves.

The Consequence:

A blanket policy of self-endorsement mathematically proves everything, including contradictions.

A formal essay cannot self-endorse. Readiness authorizes composition but does not certify correctness. The system must rely on external, human-grounded disclaimers.

Gemini Notebook

The Publication Decision Matrix

Scale: L (Low) < M (Medium) < H (High)

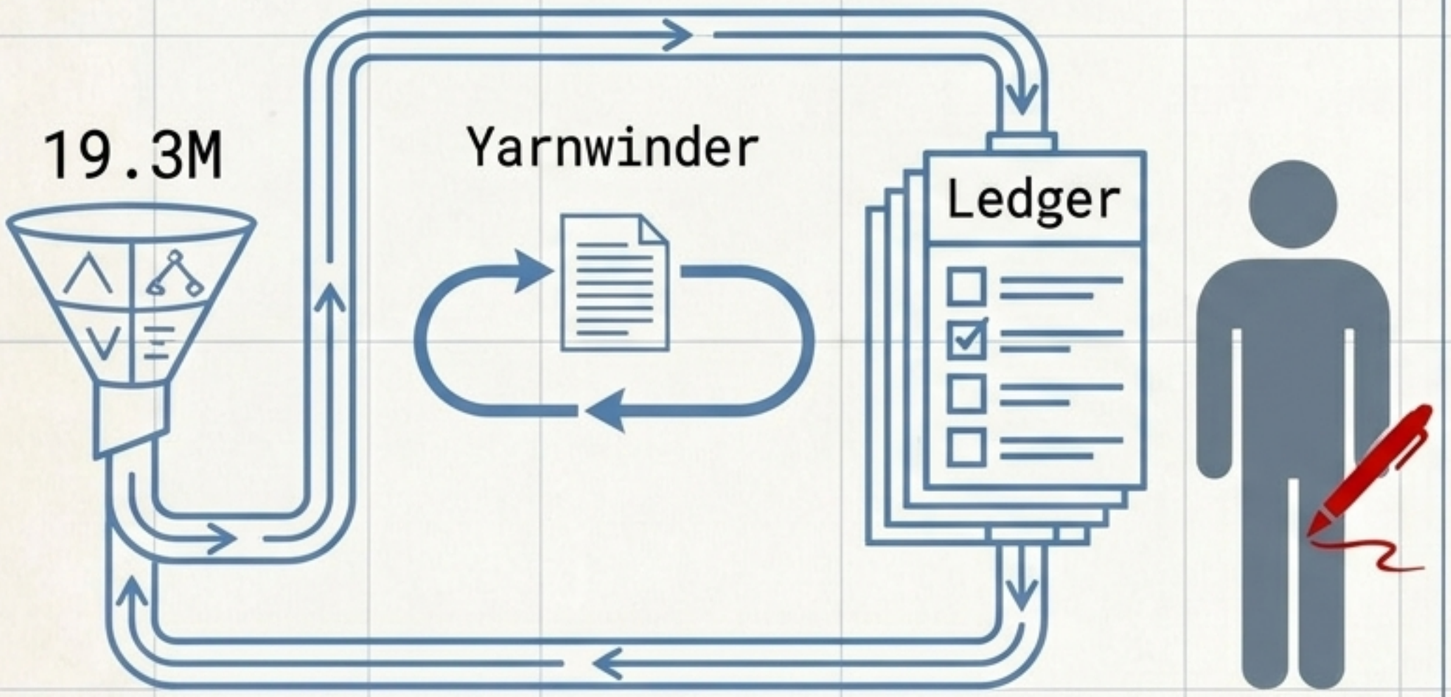
Options	Record (H is better)	Feedback (H is better)	Misreading Exposure (L is better)	Revision Exposure (L is better)
Publish Now	H	H	H	H
Publish with Status	H	H	M	H
Hold with Hash	H	L	L	L
Hold without Hash	L	L	L	L

Pareto-Maximal

The remaining choice is a value judgment outside the calculus.

Gemini Notebook

Synthesis: The Axiomatic Human



The machine can effortlessly generate 19,250,136 combinations.

It can process text, route roles, and propose connections at scale.

But the system is formally and economically empty without external intervention.

- Only a human can clear the Backlog.
- Only a human can clear the Ledger of Obligations.
- Only a human can inject the Axioms that authorize truth.

The automated system does not replace the author; it brutally distills the author's exact constraints.

Gemini Notebook